LLM-Informed Bayesian Priors

Reducing Experimental Cost in Behavioral Economics

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April 27, 2026

CSE 8803 IUQ Spring 2026 Georgia Tech

The cost problem

- A behavioral economics study costs **$10–$50 per participant**, plus IRB and lab time.
- A typical condition needs **80–200 subjects** to estimate one parameter precisely.
- An LLM, prompted with the same instructions, produces **500 simulated decisions in minutes**.

Question
How much real data does an LLM prior save, and when does it break calibration?

Setup

Data

- 137 dictator-game decisions
- von Blanckenburg et al., 2023
- 69 giving, 68 taking
- Train / test: 96 / 41 (30%)

Likelihood

$$y_i \mid \alpha, \beta \sim \mathrm{Beta}(\alpha, \beta), \quad \mu = \tfrac{\alpha}{\alpha+\beta}$$

Quantity of interest

$$\rho = \frac{n^*_{\text{LLM}}}{n^*_{\text{flat}}}$$

n^* is the smallest sample size to bring $\mathrm{Var}(\mu \mid \mathbf{y})$ below a target.

$\rho < 1 \;\Rightarrow\;$ LLM prior saves human data.

Two methods

Baseline: flat prior + MH-MCMC (log-space)

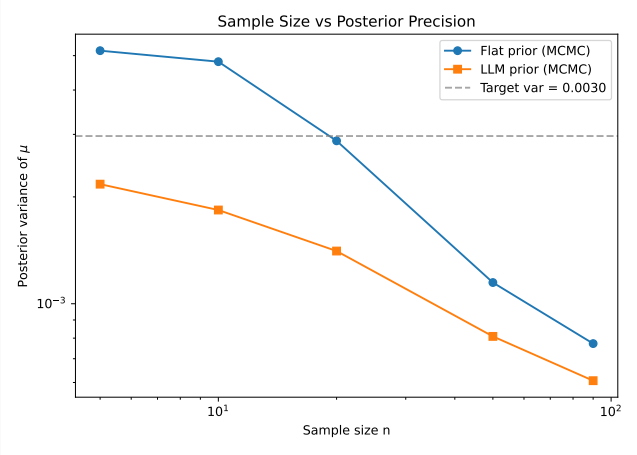
- $p(\alpha, \beta) \propto 1$ on $(0, 100)^2$
- 15k samples, 3k burn-in
- Acceptance 0.22–0.58 ✓

Frontier: LLM power prior + SVGD

- $M = 500$ pseudo-obs from GPT-4o
- $p_{\text{LLM}} \propto \prod_j \mathrm{Beta}(\tilde{y}_j; \alpha, \beta)^{w/M}$
- $w = 20$ effective weight
- SVGD: 100 particles, RBF kernel, Adam

$$\boldsymbol{\eta}_{t+1}^{(k)} = \boldsymbol{\eta}_t^{(k)} + \tfrac{\epsilon}{K} \sum_{\ell=1}^{K} \left[\kappa(\boldsymbol{\eta}_t^{(\ell)}, \boldsymbol{\eta}_t^{(k)}) \, \nabla \log p(\boldsymbol{\eta}_t^{(\ell)} \mid \mathbf{y}) + \nabla_{\boldsymbol{\eta}_t^{(\ell)}} \kappa(\boldsymbol{\eta}_t^{(\ell)}, \boldsymbol{\eta}_t^{(k)}) \right]$$

Main result: giving game



n	Var ratio
5	2.38×
10	2.62×
20	2.04×
50	1.42×
90	1.27×

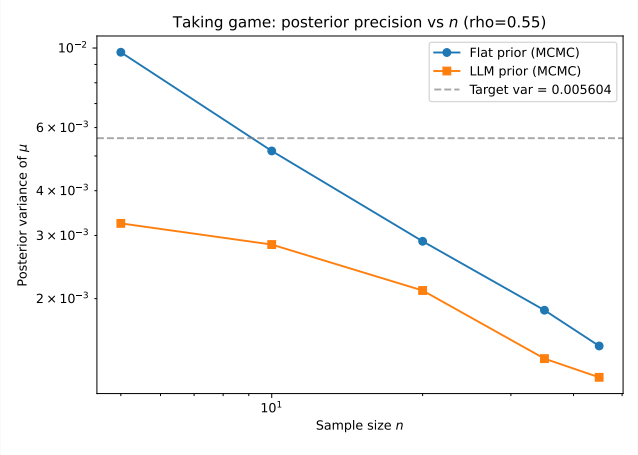
$n^*_{\text{flat}} = 19.2$

$n^*_{\text{LLM}} = 5.0$

$\rho = 0.26$

74% smaller human sample to reach the same precision.

Cross-game replication: taking game



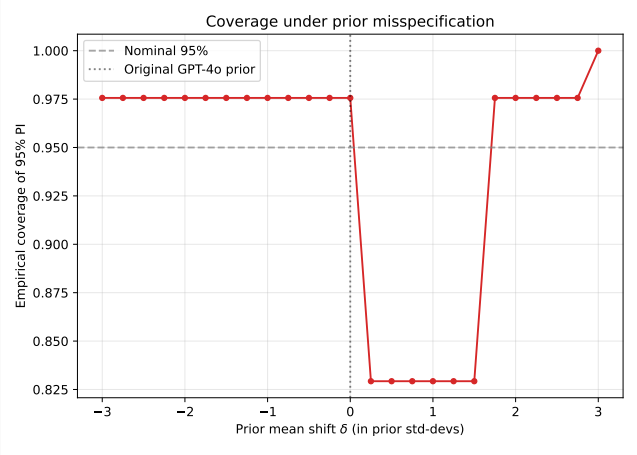
n	Var ratio
5	3.00×
10	1.82×
20	1.37×
35	1.36×
45	1.22×

$\rho_{\text{taking}} = 0.55$

vs. $\rho_{\text{giving}} = 0.26$

The LLM prior generalises – partially.

Failure mode: dense misspecification curve



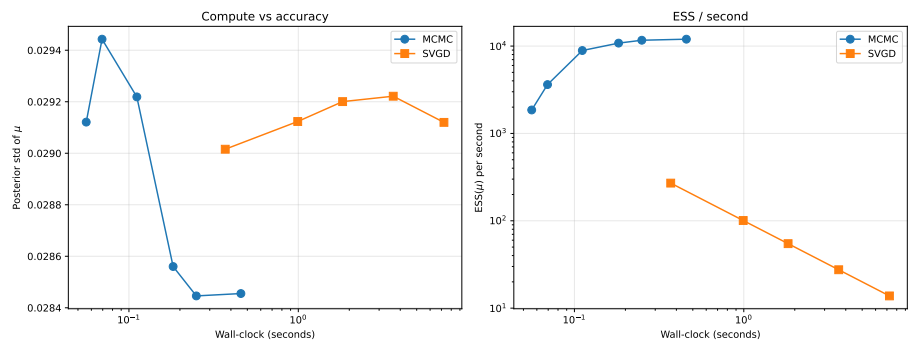
- Shift LLM prior by δ prior std-deviations
- Coverage of nominal-95% PI on held-out

Asymmetric failure:

- $\delta \leq 0$: stays calibrated
- $\delta \in [+0.25, +1.5]$: coverage drops to 0.83
- $\delta \geq +1.75$: data wins

Moderate misspecification is the dangerous regime.

Compute–accuracy: MCMC vs SVGD



MCMC at $d = 2$: ESS/sec $\approx 1.2 \times 10^4$
SVGD at $d = 2$: ESS/sec $\approx$ 14–270

For this 2-parameter Beta posterior, MCMC dominates SVGD by $\sim$100$\times$.
SVGD's value is in higher-d posteriors with multi-modality, not here.

Leakage check: gpt-3.5-turbo (Sep 2021 cutoff)

Model	Cutoff	Prior mean	ρ	Coverage ($n = 50$)
gpt-4o	2023-10	0.299	**0.261**	**0.976**
gpt-3.5-turbo	2021-09 (pre-Horton)	0.548	**0.261**	**0.829**

- **Sample-size reduction is identical**: $\rho_{\text{gpt-3.5}} = \rho_{\text{gpt-4o}}$ to 3 sig figs $\Rightarrow$ *not a leakage artefact*.
- **Calibration is not**: gpt-3.5 prior mean is 0.55, far above the human mean of 0.36.
- Power prior: variance benefit driven by w; calibration driven by prior *location*.

Limitations

Identified

- LLM under-takes vs humans in taking-game framing
- Asymmetric coverage trough at $\delta \approx +1$
- $w = 20$ is a knob, not optimised
- 2-d posterior is too small for SVGD to win
- German university student sample $\neq$ LLM persona

What's next

- Hierarchical $(\alpha_g, \beta_g, \alpha_t, \beta_t, \tau)$ joint model – SVGD becomes essential at $d \geq 4$
- Predictive log-likelihood to optimise w
- Multiple LLM corpora to disentangle leakage from prior-location quality
- Out-of-domain games where LLM training is sparse

Takeaways

- $\rho \approx 0.26$ on the giving game: GPT-4o saves $\sim$74% of the human sample.
- $\rho \approx 0.55$ on the taking game: LLM prior generalises across protocols, not framings.
- Calibration: nominal at unshifted prior; trough at moderate upward shift, not at extreme shifts.
- Leakage: not the source of the sample-size benefit, but plausibly part of why gpt-4o has a better-located prior than gpt-3.5.
- SVGD vs MCMC at $d = 2$: MCMC wins by $100\times$ in ESS/sec; SVGD's value awaits the higher-d hierarchical extension.

github.gatech.edu/bfeng66/cse8803-iuq-project

Backup: reproduce in <1 hour

```
conda activate base   # numpy / scipy / matplotlib / openai
python scripts/run_experiment.py            # rho ~ 0.26 on giving game
python scripts/run_experiment_taking.py     # rho ~ 0.55 on taking game
python scripts/run_experiment_gpt35.py      # gpt-3.5 leakage ablation
python scripts/run_failure_mode_dense.py    # 25-point misspecification grid
python scripts/run_compute_pareto.py        # MCMC vs SVGD Pareto
```

- All seeds fixed at 42
- Elicited LLM pseudo-data committed to `data/llm_prior_samples*.npy`
- API key in `.env` required only for re-elicitation